# Pythagoras Theorem

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**Written by:** Dan Finlay
**Reviewed by:** Lucy Kirkham
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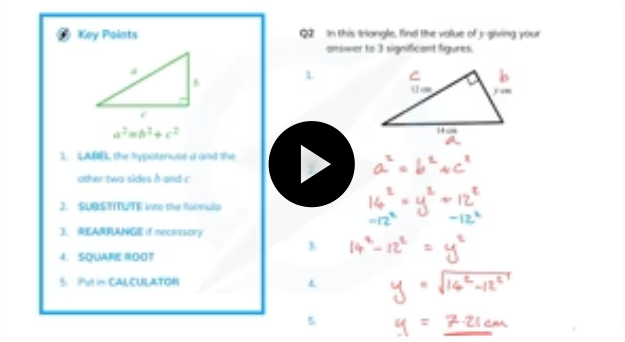


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# Pythagoras theorem

## Who is Pythagoras?

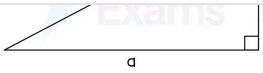
- **Pythagoras** was a **Greek mathematician** who lived over 2500 years ago
- He is most famous for **Pythagoras' theorem**, which includes the important formula for **right-angled triangles**

## What is Pythagoras' theorem?

- **Pythagoras' theorem** is a formula that links the **lengths** of the **three sides** of a **right-angled triangle**
- The **longest side** of a right-angled triangle is called the **hypotenuse**
  - The hypotenuse will always be the side **opposite** the right angle
- **Pythagoras' theorem** states that $a^2 + b^2 = c^2$
  - $c$ is the **length** of the **hypotenuse**
  - $a$ and $b$ are the **lengths** of the **two shorter sides**
    - It does **not matter** which is labelled $a$ and which is labelled $b$

Help me understand

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

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## How do I use Pythagoras' theorem to find the length of the hypotenuse?

- To find the length of the **hypotenuse**
  - **Square** the lengths of the **two shorter sides**
  - **Add** these two numbers together
  - Take the positive **square root**
- This can be written as $c = \sqrt{a^2 + b^2}$
  - This is just a **rearrangement** of the formula $a^2 + b^2 = c^2$ to make $c$ the subject
  - Note that when finding the hypotenuse you **add** inside the square root

## How do I use Pythagoras' theorem to find the length of a shorter side?

- To find the length of a **shorter side**
  - **Square** the lengths of the **hypotenuse** and the **other shorter side**
  - **Subtract** these numbers to find the difference
  - Take the positive **square root**
- This can be written as $a = \sqrt{c^2 - b^2}$
  - This is just a **rearrangement** of the formula $a^2 + b^2 = c^2$ to make $a$ the subject
  - Note that when finding one of the shorter sides you **subtract** inside the square root

## Can I use Pythagoras' theorem with other shapes?

- You can use Pythagoras' theorem with any shape that can be **split into right-angled triangles**
- You can find the **length** of the **diagonal** of a **rectangle** or **square**
  - Use its diagonal to split the shape into two identical right-angled triangles
  - The diagonal is now the hypotenuse of the right-angled triangles

### Examiner Tips and Tricks

If the hypotenuse ends up being shorter than another side in your answer then you have made a **mistake** somewhere.

Make sure that you subtract the **smaller value from the bigger value** when finding the length of a shorter side.

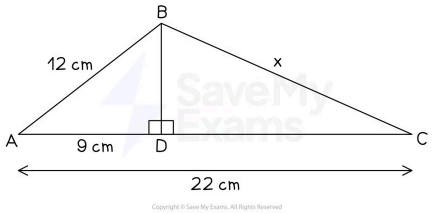
## Examiner Tips and Tricks

In questions with **multiple steps**:

- Leave your answer as an **exact answer**
- **Do not round** until the very **end of the question**

## Worked Example

In the following diagram:
$AB = 12 \text{ cm}$
$AC$ is a straight line, with $AD = 9 \text{ cm}$ and $AC = 22 \text{ cm}$




Find $x$, the length of side $BC$. Give your answer to 1 decimal place.

**Answer:**

To find $x$, we first need to find the length of $BD$ using triangle $ABD$
Note that $BD$ is a shorter side
Apply Pythagoras' theorem, $a = \sqrt{c^2 - b^2}$

$$BD = \sqrt{12^2 - 9^2} = \sqrt{63} = 7.93725...$$

It is best to leave rounding until the very end, use $\sqrt{63}$ (or $3\sqrt{7}$ if this is what your calculator has given you) in subsequent working

Find the length of $DC$ by subtracting the length of $AD$ from the length of $AC$

$$DC = 22 - 9 = 13 \text{ cm}$$

Now we can find $x$ using triangle $BCD$
Note that $BC$ is the hypotenuse
Apply Pythagoras' theorem, $c = \sqrt{a^2 + b^2}$

$$x = \sqrt{BD^2 + DC^2} = \sqrt{(\sqrt{63})^2 + 13^2} = \sqrt{63 + 169}$$

$$x = \sqrt{232} = 15.23154621...$$

Round to 1 decimal place

$$x = 15.2 \text{ cm}$$

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